L2.7 Newton's Method

§3.8 — when no formula will give you the root

WHERE ROOT-FINDING RUNS OUT

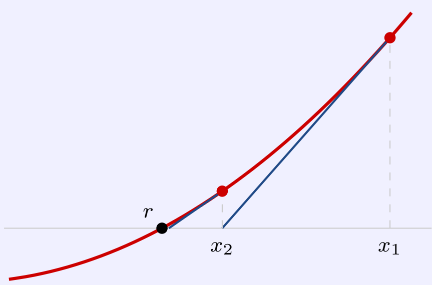
degree 2	**Loh.** Always works, always fast
degree 3, 4	a formula exists and is unusable. The **Rational Root Theorem** with the **Remainder Theorem** *can* work — but only ever finds *rational* roots
degree 5 and up	**no formula at all.** Not a complicated one — none

And it is not only about degree: $\cos x = x$ has an ordinary solution near 0.739 with no polynomial in sight.

"an irrational root is invisible to the Rational Root Theorem no matter how long the list gets"

Newton's method

The tangent line hugs the curve near the point of tangency, so **where the tangent crosses the axis is close to where the curve crosses it** — and the tangent's crossing you can solve for exactly. That is linearization, pointed at the x-axis.



Take the tangent at $\big(x_1, f(x_1)\big)$, ask where its height is 0, and call that x_2. Then do it again to x_2, and again.

The division needs $f'(x_1) \neq 0$ — a nearly flat tangent is what makes the method fail.

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad \Longleftrightarrow \qquad \lim_{n\to\infty} x_n = r}$$

The left box is the **procedure**. The right box is the **claim it makes**: the approximations close in on r — not that any one of them lands on it.

"x_1 = 1st guess · x_2 = second guess = closer to root · x_3 = repeat Newton's Method ⇒ even closer to root"

A number you want, turned into an equation you can iterate. $\sqrt[6]{2}$ is the positive root of $x^6 - 2 = 0$. That first line is the whole move, and it works for any root of any number.

When to stop. The method never announces it is finished. Stop when x_n and x_{n+1} **agree to the number of decimal places you were asked for.** It is a rule of thumb, not a theorem — which is why the answer is written with $\approx$ and never with $=$.

CHOOSING x_1

1. Use the Intermediate Value Theorem to bracket the root. If f is continuous and changes sign between two numbers, a root lives between them. For $x^3 - 2x - 5$: $f(1) = -6$, $f(2) = -1$, $f(3) = 16$ — so a root lives in $[2, 3]$ and $x_1 = 2$ is a *reading*, not a guess.

2. Look at the graph. Avoid a critical number — a local min or Max, where $f' = 0$ — sitting between x_1 and the root. The tangent near it is nearly flat. Move x_1 to the same side of it as the root.

"a few substitutions before you start are worth more than three extra iterations after"

WATCH OUT

✗ ~~Newton's method finds the root~~ → **it produces a sequence closing in on one. You stop when the digits you need stop changing**

✗ ~~subtract f/f' once and you are done~~ → **iterate. You use the formula again on the answer it just gave you**

✗ ~~x_2 is always closer to r than x_1~~ → **not if the tangent is nearly flat. Look at $f'(x_1)$ before you trust the step**

✗ ~~a nearly flat tangent is the only way it fails~~ → **it is the most common way — but $\sqrt[3]{x}$ fails from every starting value. There the trouble is the function, not the choice**

✗ ~~a bad first guess just takes longer~~ → **it can land on a different root, leave the domain, or never converge at all**

✗ ~~the equation has to be a polynomial~~ → **any f you can differentiate. $\sin x - x^2$ works, and nothing in the derivation mentioned degree**

✗ ~~there is no way to get $\sqrt[6]{2}$~~ → **turn the number into an equation. It is the positive root of $x^6 - 2 = 0$**